

Group 2

Predicting Accounting Fraud

Team Name: XXX

Individual Names: XXX

We estimate the following fraud prediction model:

$$\text{Fraud}_t = \alpha + \theta_1 \text{M-Flag} + \theta_2 \text{F-Flag} + \beta_1 Z_1 + \beta_2 Z_2 + \cdots + \beta_n Z_n + \varepsilon \quad (1)$$

Fraud_t is an indicator variable equal to one if the financial results for the firm-year were the subject of an SEC Accounting and Auditing Enforcement Release (AAER), and zero otherwise. M-Flag is the flag for fraud using the Beneish M-score (M-score > -1.78). F-Flag is the flag for fraud using the Dechow F-score (F-score > 1.0). Z 's are additional predictor variables described below. Table 1 provides variable definitions.

Following Beneish (1999), Beneish, Lee, and Nichols (2011), and Dechow et al. (2011), we examine the M-Flag and F-Flag variables as required predictors of fraud. We supplement these with three additional variables — ACCRUAL, ASSET_GROWTH, and LEV — each motivated by a distinct fraud incentive channel. Note: ROA ($= IB/AT$) was initially included but dropped due to near-perfect collinearity with ACCRUAL ($r = 0.905$, as both contain IB/AT). SALES_GROWTH was dropped due to extreme outliers (std = 61, max = 11,880) that rendered it uninformative. Table 2 summarizes the descriptive statistics.

Table 3 provides regression results. t -statistics are reported in parentheses. *, **, *** denote coefficients significantly different from zero at the 10%, 5%, and 1% level respectively. Cells with significant coefficients are shaded green.

Findings: Consistent with our predictions, we find that F-Flag is positive and highly significant ($\hat{\theta}_2 = 0.0073$, $t = 8.31$), indicating that firms flagged by the Dechow F-Score are approximately 0.73 percentage points more likely to commit fraud — a near-doubling of the unconditional fraud rate of 0.70%. M-Flag is positive but statistically insignificant ($t = 0.22$), consistent with the finding in Dechow et al. (2011) that the F-Score has greater discriminatory power than the M-Score for predicting AAER firms. Among our additional variables, ACCRUAL and LEV are in the predicted direction but insignificant, while ASSET_GROWTH is marginally positive ($t = 1.61$, $p = 0.11$). The low R^2 (0.0018) is expected given the rarity of fraud ($< 1\%$ of firm-years) and is consistent with values reported in the literature.

Table 1: Variable Definitions

Variable
DEPENDENT VARIABLE
Fraud_t
INDEPENDENT VARIABLES
M-FLAG
F-FLAG
ACCRUAL
ASSET_GROWTH
LEV

Findings: Results from estimating the regression by year indicate that F-Flag is the most consistent predictor across the sample period, achieving significance at the 5% level or better in 8 of 12 years (2003–2006, 2011, 2012, 2014) and at the 10% level in 2007. Its coefficient is positive in every year, ranging from 0.0013 in 2008 to 0.0210 in 2003. M-Flag is never significant in any individual year and alternates in sign, confirming its limited incremental predictive power conditional on F-Flag. The 2008–2010 period shows no significant predictors, consistent with reduced AAER enforcement activity during and after the financial crisis, as evidenced by the decline in fraud rates from 1.57% in 2003 to 0.44% in 2008. ASSET_GROWTH achieves significance in 2005 and 2014, consistent with the prediction that aggressive asset expansion is episodically associated with fraudulent reporting during periods of elevated capital market

Table 2: Descriptive Statistics

Variable	Obs	Mean	Std	25th	Median	75th	Years
Fraud _t	41,959	0.007	0.085	0.000	0.000	0.000	2003–2014
M-Flag	41,959	0.121	0.326	0.000	0.000	0.000	2003–2014
F-Flag	41,959	0.348	0.476	0.000	0.000	1.000	2003–2014
ACCRUAL	41,959	−0.079	0.508	−0.099	−0.053	−0.019	2003–2014
ASSET_GROWTH	41,959	0.126	0.646	−0.039	0.051	0.166	2003–2014
LEV	41,959	0.417	0.405	0.240	0.388	0.539	2003–2014

All variables defined in Table 1. Sample: fiscal years 2003–2014, requiring positive *AT*, *SALE*, *PPENT* and non-missing values for all model variables.

Table 3: Regression Results — All Years

Independent Variable	ALL YEARS Fraud _t
Intercept	0.0053*** (7.22)
M-Flag	0.0003 (0.22)
F-Flag	0.0073*** (8.31)
ACCRUAL	−0.0007 (−0.59)
ASSET_GROWTH	0.0011 (1.61)
LEV	−0.0021 (−1.45)
Observations	41,959

OLS estimates. Dependent variable: Fraud_t. *t*-statistics in parentheses. **p* < 0.10, ***p* < 0.05, ****p* < 0.01. Shaded cells are significant at 10% or better.

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Table 4: Regression Results by Year — 2003 to 2008

Independent Variable	2003 Fraud _t	2004 Fraud _t	2005 Fraud _t	2006 Fraud _t	2007 Fraud _t	2008 Fraud _t
Intercept	0.0123*** (3.43)	0.0116*** (3.30)	0.0116*** (3.27)	0.0054* (1.94)	0.0068** (2.52)	0.0066*** (3.05)
M-Flag	-0.0059 (-0.99)	-0.0011 (-0.21)	0.0003 (0.05)	0.0006 (0.14)	-0.0008 (-0.20)	0.0005 (0.13)
F-Flag	0.0210*** (4.91)	0.0109*** (2.98)	0.0101*** (2.82)	0.0089*** (3.22)	0.0048* (1.83)	0.0013 (0.54)
ACCRUAL	0.0024 (0.21)	0.0037 (0.50)	-0.0009 (-0.07)	0.0001 (0.03)	0.0075 (0.91)	0.0031 (0.71)
ASSET_GROWTH	0.0007 (0.30)	0.0009 (0.29)	0.0078** (2.04)	0.0049 (1.59)	0.0003 (0.15)	0.0011 (0.40)
LEV	-0.0053 (-0.78)	-0.0077 (-1.16)	-0.0139** (-1.99)	-0.0077 (-1.42)	-0.0058 (-1.15)	-0.0053 (-1.41)
Observations	4,152	3,996	3,803	3,721	3,529	3,448

OLS estimates by fiscal year. Dependent variable: Fraud_t. *t*-statistics in parentheses.

p* < 0.10, *p* < 0.05, ****p* < 0.01. Shaded cells are significant at 10% or better.

Appendix A

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1  /*=====
2  ACCT 270 - Group Assignment 2: Predicting Accounting Fraud
3  Model: Fraud = a + b1*MFLAG + b2*FFLAG + b3*ACCRUAL
4          + b4*ASSET_GROWTH + b5*LEV + e
5  Sample: 2003-2014 | Data: Compustat, AAER
6  =====*/
7
8  libname mydata "s:\sasdata";
9  libname home "d:\users\mcba\desktop\programs";
10 options ps=max ls=max nocenter;
11
12 /*-----
13 STEP 1: CONSTRUCT M-SCORE (Beneish 1999)
14 MSCORE = -4.84 + 0.92*DSR + 0.528*GMI + 0.404*AQI + 0.892*SGI
15          + 0.115*DEPI - 0.172*SGAI + 4.679*ACCRUALS - 0.327*LEVI
16 MFLAG = 1 if MSCORE > -1.78
17 -----*/
18
19 /* Examine AAER dataset to confirm structure and year coverage */
20 proc contents data=mydata.aaer1982to2014; run;
21 proc freq data=mydata.aaer1982to2014; tables fyear; run;

```

Table 5: Regression Results by Year — 2009 to 2014

Independent Variable	2009 Fraud _t	2010 Fraud _t	2011 Fraud _t	2012 Fraud _t	2013 Fraud _t	2014 Fraud _t
Intercept	0.0046* (1.83)	0.0045* (1.95)	0.0015 (0.59)	−0.0002 (−0.06)	0.0010 (0.53)	−0.0010 (−0.65)
M-Flag	0.0005 (0.12)	0.0011 (0.26)	−0.0056 (−1.36)	−0.0011 (−0.24)	0.0005 (0.14)	0.0037 (1.46)
F-Flag	0.0025 (0.83)	0.0018 (0.63)	0.0055** (2.13)	0.0097*** (3.39)	0.0024 (1.08)	0.0044*** (2.88)
ACCRUAL	−0.0011 (−0.15)	0.0010 (0.32)	0.0036 (0.40)	−0.0002 (−0.03)	−0.0009 (−0.15)	−0.0032 (−0.61)
ASSET_GROWTH	0.0003 (0.25)	0.0008 (0.48)	−0.0001 (−0.03)	−0.0031 (−1.28)	−0.0006 (−0.37)	0.0035** (2.25)
LEV	0.0005 (0.09)	0.0013 (0.29)	0.0049 (0.97)	0.0071 (1.27)	0.0032 (0.90)	0.0001 (0.03)
Observations	3,449	3,297	3,178	3,166	3,131	3,089

OLS estimates by fiscal year. Dependent variable: Fraud_t. *t*-statistics in parentheses.

p* < 0.10, *p* < 0.05, ****p* < 0.01. Shaded cells are significant at 10% or better.

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22
23 /* Pull Compustat; require positive AT, SALE, PPENT and
24    non-missing core variables for M-score and F-score          */
25 data variables;
26    set mydata.compustat1990to2019;
27    if AT > 0 and SALE > 0 and PPENT > 0;
28    if nmiss(cogs, dp, rect, act, lct, ib, oancf) = 0;
29
30    /* Fill optional variables with zero if missing */
31    if dlтт = . then dlтт = 0;
32    if xsга = . then xsга = 0;
33    if am   = . then am   = 0;
34    if invт = . then invт = 0;
35    if che  = . then che  = 0;
36    if dltis= . then dltis= 0;
37    if sstk = . then sstk = 0;
38
39    /* M-Score raw components */
40    DS      = RECT / SALE;
41    GM      = 1 - (COGS / SALE);
42    AQ      = 1 - ((PPENT + ACT) / AT);
43    SG      = SALE;
44    DEP     = (DP - AM) / (DP - AM + PPENT);

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45     SGA      = XSGA / SALE;
46     ACCRUAL = (IB - OANCF) / AT;          /* Also used as Z1 predictor */
47     LEV      = (LCT + DLTT) / AT;        /* Also used as Z3 predictor */
48
49     /* ROA computed but excluded from regression:
50        near-perfect collinearity with ACCRUAL (r=0.905) */
51     ROA = IB / AT;
52 run;
53
54 /* Self-join to prior year for M-score index ratios,
55    lagged asset/sale values for growth variables          */
56 proc sql;
57     create table addlagyear as
58     select a.*,
59           b.DS as DS_m1, b.GM as GM_m1, b.AQ as AQ_m1,
60           b.SG as SG_m1, b.DEP as DEP_m1, b.SGA as SGA_m1,
61           b.ACCRUAL as ACCRUAL_m1, b.LEV as LEV_m1,
62           b.AT as AT_m1, b.IB as IB_m1,
63           b.RECT as RECT_m1, b.INVT as INVT_m1,
64           b.SALE as SALE_m1
65     from variables a
66     inner join variables b
67         on a.gvkey = b.gvkey and a.fyear - 1 = b.fyear;
68 quit;
69
70 /* Compute M-Score indices (set to 1 if denominator = 0),
71    MSCORE, MFLAG, and additional predictor variables          */
72 data Mscore;
73     set addlagyear;
74     if DS_m1 > 0 then DSR = DS / DS_m1; else DSR = 1;
75     if GM      > 0 then GMI = GM_m1 / GM; else GMI = 1;
76     if AQ_m1 > 0 then AQI = AQ / AQ_m1; else AQI = 1;
77     if SG_m1 > 0 then SGI = SG / SG_m1; else SGI = 1;
78     if DEP      > 0 then DEPI = DEP_m1 / DEP; else DEPI = 1;
79     if SGA_m1 > 0 then SGAI = SGA / SGA_m1; else SGAI = 1;
80     if LEV_m1 > 0 then LEVI = LEV / LEV_m1; else LEVI = 1;
81
82     MSCORE = -4.84 + 0.92*DSR + 0.528*GMI + 0.404*AQI + 0.892*SGI
83             + 0.115*DEPI - 0.172*SGAI + 4.679*ACCRUAL - 0.327*LEVI;
84     if MSCORE > -1.78 then MFLAG = 1; else MFLAG = 0;
85
86     /* Z2: asset growth using lagged total assets */
87     if AT_m1 > 0 then ASSET_GROWTH = (AT - AT_m1) / AT_m1;
88     /* SALES_GROWTH dropped: extreme outliers (std=61, max=11880) */
89     if SALE_m1 > 0 then SALES_GROWTH = (SALE - SALE_m1) / SALE_m1;
90 run;
91

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92  /*-----
93  STEP 2: CONSTRUCT F-SCORE (Dechow et al. 2011)
94  Requires 2-year lagged receivables for CH_CS
95  FFLAG = 1 if FSCORE > 1.0
96  -----*/
97
98  /* Get 2-year lagged receivables from raw Compustat */
99  proc sql;
100      create table addlagyear2 as
101      select a.*, b.rect as rect_m2
102      from Mscore a
103      inner join mydata.compustat1990to2019 b
104              on a.gvkey = b.gvkey and a.fyear - 2 = b.fyear;
105  quit;
106
107  /* Compute 7 F-Score components, log-odds, probability, F-Score */
108  data Fscore;
109      set addlagyear2;
110      if nmiss(rect,rect_m1,rect_m2,ib,ib_m1,invnt,invnt_m1,
111              che,at,at_m1) = 0;
112
113      RSST    = ACCRUAL;
114      CH_REC  = (RECT - RECT_m1) / AT;
115      CH_INV  = (INVT - INVT_m1) / AT;
116      SOFT    = (AT - PPENT - CHE) / AT;
117      CH_ROA  = (IB/AT) - (IB_m1/AT_m1);
118
119      /* If denominator of CH_CS is zero, set to 1 */
120      if SALE_m1 - (RECT_m1 - RECT_m2) > 0
121          then CH_CS = ((SALE-(RECT-RECT_m1)) /
122                      (SALE_m1-(RECT_m1-RECT_m2))) - 1;
123      else CH_CS = 1;
124
125      if DLTIS + SSTK > 0 then ISSUE = 1; else ISSUE = 0;
126
127      PredictedValue = -7.893 + 0.790*RSST + 2.518*CH_REC
128                      + 1.191*CH_INV + 1.979*SOFT
129                      + 0.171*CH_CS - 0.932*CH_ROA + 1.029*ISSUE;
130      /* Cap at 709 to prevent exp() overflow */
131      if PredictedValue > 709 then PredictedValue = 709;
132
133      Probability = exp(PredictedValue) / (1 + exp(PredictedValue));
134      Fscore      = Probability / 0.0037;
135      if FSCORE > 1 then FFLAG = 1; else FFLAG = 0;
136  run;
137
138  /*-----

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139 STEP 3: MERGE AAER, CODE FRAUD, RESTRICT TO 2003-2014
140 -----*/
141
142 /* Left join: keep all firm-years; match to AAER => fraud=1 */
143 proc sql;
144     create table addAAER as
145     select a.*, b.P_aaer
146     from FSCORE a
147     left join mydata.aaer1982to2014 b
148         on a.gvkey = b.gvkey and a.fyear = b.fyear;
149 quit;
150
151 data Fraud;
152     set addAAER;
153     if 2003 <= fyear <= 2014;
154     if nmiss(mflag, fflag) = 0;
155     if nmiss(accrual, asset_growth, lev) = 0;
156     if not missing(P_aaer) then FRAUD = 1; else FRAUD = 0;
157 run;
158
159 /*-----*/
160 STEP 4: DESCRIPTIVE STATISTICS
161 -----*/
162
163 proc means data=fraud n mean std p25 p50 p75 maxdec=3;
164     var fraud mflag fflag accrual asset_growth lev;
165 quit;
166
167 proc freq data=fraud; tables fyear; run;
168
169 proc means data=fraud mean maxdec=4;
170     var fraud; class fyear;
171 run;
172
173 proc corr data=fraud;
174     var mflag fflag accrual asset_growth lev;
175 quit;
176
177 /*-----*/
178 STEP 5: PART I - POOLED REGRESSION (2003-2014)
179 -----*/
180
181 proc reg data=fraud;
182     model fraud = mflag fflag accrual asset_growth lev;
183     title 'Part I: Pooled Fraud Regression (2003-2014)';
184 quit;
185

```



```
186 /*-----  
187 STEP 6: PART II - YEAR-BY-YEAR REGRESSIONS  
188 BY statement loops over sorted fyear values  
189 -----*/  
190  
191 proc sort data=fraud; by fyear; run;  
192  
193 proc reg data=fraud;  
194     by fyear;  
195     model fraud = mflag fflag accrual asset_growth lev;  
196     title 'Part II: Year-by-Year Fraud Regressions';  
197 quit;
```